LESSON 3: UNBOUNDED OPERATORS and SPECTRUM AND RESOLVENT
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## 1. Unbounded operators

The notion of spectrum for linear operators:

- The spectrum serves as the infinite-dimensional equivalent of matrix eigenvalues.
- Many fundamental results from matrix theory can be adapted to linear operators $A$ acting on a Banach or Hilbert space.
- Bounded (continuous) linear operators are often not suitable for several applications in Mathematical Physics.
- We also consider the set of unbounded (discontinuous) operators, in particular those operators called "closed" operators.

**Definition 1.1** (Definition of Unbounded Linear Operator)**.** *An unbounded linear operator from Banach spaces $X$ to $Y$ is a pair $(A, D(A))$ where*

- *$D(A) \subset X$ is a linear subspace,*
- *$A : D(A) \to Y$ is a linear map, i.e.,*
  (1) ***Additivity**: For all $x, y \in D(A)$,*
  $$A(x + y) = A(x) + A(y).$$
  (2) ***Homogeneity**: For all $x \in D(A)$ and all scalars $\alpha \in \mathbb{F}$ (where $\mathbb{F}$ is typically $\mathbb{R}$ or $\mathbb{C}$),*
  $$A(\alpha x) = \alpha A(x).$$

We motivate the definition of operators defined on domains with a simple example using multiplication operators.

**Example 1.2** (Why We Need Domains for Certain Operators)**.** *We have that*

$$h_\alpha(x) = x^{-\alpha} \in L^2(0, 1) \text{ if and only if } \alpha < \frac{1}{2}.$$

*So take*

$$\mu(x) = \frac{1}{x^{3/8}} \quad \text{and} \quad f(x) = \frac{1}{x^{1/4}} \in L^2(0, 1), \quad \text{but} \quad \mu f \notin L^2(0, 1)$$

*since*

$$\mu(x)f(x) = \frac{1}{x^{3/8}} \cdot \frac{1}{x^{1/4}} = \frac{1}{x^{5/8}} \quad \text{with} \quad \frac{5}{8} > \frac{1}{2}$$

*So if we define the operator*

$$A_\mu : L^2(0, 1) \to L^2(0, 1), \quad A_\mu(f) := \mu(x)f(x)$$

*$f$ cannot belong to the entire space $L^2(0, 1)$ but it has to belong to the subspace*

$$D(A_\mu) = \{f \in L^2(0, 1) \ : \ \mu f \in L^2(0, 1)\}$$

*That is, we take those $f \in L^2(0, 1)$ such that $A_\mu(f) \in L^2(0, 1)$.*

**Definition 1.3** (densely defined operator)**.** *A linear operator is densely defined if $D(A)$ is dense in $X$.*

**Remark 1.4.** *We will say "linear operator" (or simply "operator") for an unbounded linear operator when we specify the domain, since in this case no confusion will arise.*

**Definition 1.5** (Kernel of a Linear Operator)**.** *Let $A : D(A) \subset X \to Y$ be a linear operator from $D(A)$ to a vector space $Y$. The kernel (or null space) of $A$, denoted by $\ker(A)$, is the linear subspace:*

$$\ker(A) = \{x \in D(A) : A(x) = 0_Y\}$$

*where $0_Y$ is the zero vector in $Y$.*

**Definition 1.6** (Range of a Linear Operator)**.** *Let $A : D(A) \subset X \to Y$ be a linear operator from $D(A)$ to a vector space $Y$. The range, or image, of $A$, denoted by range$(A)$ or Im$(A)$, is the linear subspace:*

$$range(A) = \{A(x) \in Y : x \in D(A)\}$$

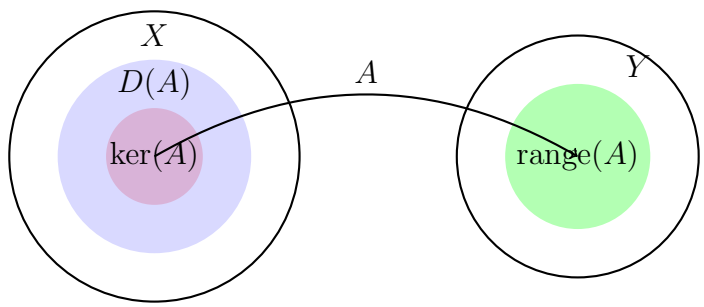



Fig. Visualization of the kernel and of the range of a linear operator.

**Definition 1.7** (Definition of Extension)**.** *We say that the operator $(B, D(B))$ is an extension of $(A, D(A))$ if*

$$D(A) \subset D(B) \quad \text{and} \quad Bx = Ax \quad \text{for all} \quad x \in D(A).$$

*We write $A \subset B$.*

**Definition 1.8** (Restriction of Operators)**.** *The restriction of $A$ to a linear subspace $D_0$ of $D(A)$ is the operator $A|_{D_0}$ with domain $D_0$ such that*

$$(A|_{D_0})x = Ax \quad \text{for all} \quad x \in D_0.$$

**Definition 1.9** (Equality Between Operators)**.** *If $A$ and $B$ are two operators from $X$ to $Y$, by definition,*

$$A = B$$

*if and only if*

$$D(A) = D(B) \quad \text{and} \quad Ax = Bx \quad \text{for all} \quad x \in D(A) = D(B).$$

**Lemma 1.10** (Associativity and distributive law for unbounded operators)**.** *Considering the unbounded operators $A$, $B$ and $C$ with their domains:*

- ***Associativity (This is still true)**: $A \cdot (B \cdot C) = (A \cdot B) \cdot C$*

- ***Distributivity (These are true in the following sense):***
$$A \cdot (B + C) \supset A \cdot B + A \cdot C$$
*and*
$$(A + B) \cdot C = A \cdot C + B \cdot C$$

## 2. The inverse of an operator

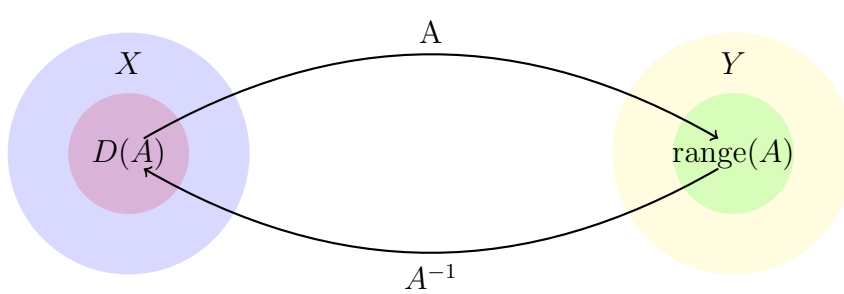
We recall the important concepts:

- **Injectivity (or one-to-one)** $A : D(A) \subset X \to Y$ is said injective (or one-to-one) if
$$\text{for any } \; x, y \in D(A) \; \text{ such that } \; x \neq y \quad \Rightarrow \quad Ax \neq Ay$$
or equivalently
$$\text{for any } \; x, y \in D(A) \; \text{ such that } \; Ax = Ay \quad \Rightarrow \quad x = y$$
- **Surjective (or onto)** $A : D(A) \subset X \to Y$ is said surjective (or onto) if $\text{range}(A) = Y$. Equivalently
$$\text{for every } \; y \in Y \; \text{ exists at least one } \; x \in D(A) \; \text{ with } \; Ax = y.$$
- **Bijective** $A$ is sad to be bijective if it is both injective and surjective.

**Definition 2.1** (Definition of the Inverse Operator)**.** *We have the following*

- *Let $A : D(A) \subseteq X \to Y$ be injective (or one-to-one). In this case, there exists the operator*
$$A^{-1} : \text{range}(A) \to D(A)$$
*with*
$$D(A^{-1}) = \text{range}(A) \quad \textit{and} \quad \text{range}(A^{-1}) = D(A).$$
- *Observe that $A : D(A) \subseteq X \to \text{range}(A)$ is surjective (or onto).*
- *Thus, $A : D(A) \subseteq X \to \text{range}(A)$ is a bijection.*

*The operator $A^{-1} : \text{range}(A) \to D(A)$ is called the inverse of $A$ and we have*
$$A^{-1}Ax = x \quad \textit{for all } x \in D(A),$$
$$AA^{-1}y = y \quad \textit{for all } y \in \text{range}(A).$$




**Theorem 2.2** (Characterization of the existence of the inverse operator with the Kernel)**.** *Let $X$ and $Y$ be Banach spaces over $\mathbb{F}$. Let $A : D(A) \subset X \to Y$ with* $\text{range}(A) \subset Y$*. Then*

- *The inverse operator*
$$A^{-1} : \mathrm{range}(A) \to D(A)$$
*exists if and only if*
$$\ker(A) = \{0\}.$$
- *If $A^{-1}$ exists, it is a linear operator.*

**Theorem 2.3.** *If two operators $A$ and $B$ are such that*
$$\ker(A) = 0 \quad \textit{and} \quad \ker(B) = 0$$
*then*
$$\ker(AB) = 0$$
*and*
$$(AB)^{-1} = B^{-1}A^{-1}.$$

**Remark 2.4** (We recall the definition of Bounded Operators)**.** *A densely defined operator is called bounded if there exists $M \geq 0$ such that*
$$||Ax||_Y \leq M||x||_X$$
*for all $x \in D(A)$.*

- *Recall that $A$ is bounded iff $A$ is continuous.*
- *Recall that denote by $\mathcal{B}(X)$ the space of all bounded linear operators from the Banach space $X$ into itself.*

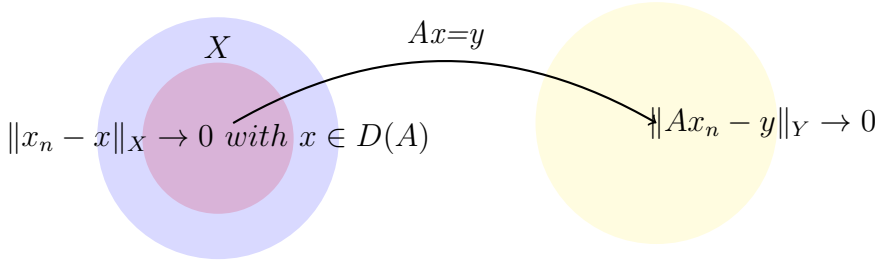
**Proposition 2.5** (Important fact)**.** *If $(A, D(A))$ is densely defined and bounded there exists a unique continuous extension $\overline{A}$ with $D(\overline{A}) = X$.*

## 3. Closed operators

**Definition 3.1** (with sequences of Closed Operators)**.** *An operator (A, D(A)) is called* ***closed*** *if the following holds: For any sequence $(x_n)_{n\in\mathbb{N}}$ in $D(A)$ such that*

- $x_n \to x$ in $X$ *and*
- $Ax_n \to y$ in $Y$,

*it holds that $x \in D(A)$ and $Ax = y$.*



**Example 3.2** (The derivative as an unbounded operator)**.** *We recall the following:*

- *Let $X = C([0,1])$ be endowed with natural norm $||\cdot||_\infty$ and consider the operator*
$$Af := f' \quad \text{with domain} \quad D(A) = C^1([0,1]).$$
*Then $A$ is linear because the derivative is linear, but it is not bounded.*
- *In fact, consider the functions*
$$u_n \in D(A) \quad \text{given by } u_n(t) = \frac{1}{\sqrt{n}} sin(nt)$$
*for $n \in \mathbb{N}$. They satisfy $||u_n||_\infty \to 0$ and*
$$\|Au_n\|_\infty = \|\sqrt{n}cos(nt)\|_\infty \geq \sqrt{n} \to \infty$$
*as $n \to \infty$. So $A$ is unbounded.*

The motivation for the definition of closed operators is suggested by differential operators.

**Theorem 3.3** (Derivative of a Sequence of Functions)**.** *Let $(f_n)$ be a sequence of functions defined on a bounded interval $I \subset \mathbb{R}$, and let $f : I \to \mathbb{R}$ be a function such that:*

- *$f_n \to f$ pointwise on $I$ as $n \to \infty$,*
- *$f_n'$ exists for all $n$ in $I$.*
- *The sequence $(f_n')$ converges uniformly to a function $g$ on $I$.*

*Then, the function $f$ is differentiable on $I$, and:*
$$f' = g.$$

**Remark 3.4.** *We motivate the definition of Closed Operators:*

*(I) We remark that if $f_n \in D(A) = C^1([0,1])$ and they satisfy $f_n \to f$ and $Af_n = f_n' \to g$ in $C([0,1])$ as $n \to \infty$, then $f \in D(A)$ and $Af = g$.*

*(II) Let $A$ be a linear operator from $X$ to $Y$. The operator $A$ is called closed if for all $x_n \in D(A)$, $n \in \mathbb{N}$, possessing limits*

- *$x = \lim_{n\to\infty} x_n$ in $X$ and*
- *$y = \lim_{n\to\infty} Ax_n$ in $Y$,*

*we have $x \in D(A)$ and $Ax = y$.*

*(III) For closed operator $A$ we thus have*
$$\lim_{n\to\infty} Ax_n = A(\lim_{n\to\infty} x_n)$$
*if $(x_n)$ and $(Ax_n)$ converge.*

**Remark 3.5.** *There is an alternative definition of closed operator.*

**Definition 3.6** (Closed Operators: Definition with graph)**.** *An operator $A$, $D(A)$ is called* ***closed*** *if its graph*

$$G(A) := \{(x, Ax) : x \in D(A)\} \subset X \times Y$$

*is closed in $X \times Y$.*

**Theorem 3.7** (Equivalent Definitions of Closed Operators)**.** *Let $(A, D(A))$ be an operator. The following statements are equivalent:*

(1) *The graph of $A$, denoted by $G(A)$, is closed in $X \times Y$.*
(2) *For any sequence $(x_n)_{n\in\mathbb{N}}$ in $D(A)$ such that $x_n \to x$ in $X$ and $Ax_n \to y$ in $Y$, it holds that $x \in D(A)$ and $Ax = y$.*

**Theorem 3.8** (Closed Graph Theorem (important theorem))**.** *Let $X$ and $Y$ be Banach spaces and let $A$ be a linear operator from $X$ into $Y$. Then $A$ is bounded if and only if the graph of $A$, defined as*

$$G(A) = \{(x, A(x)) \in X \times Y : x \in X\},$$

*is closed in $X \times Y$.*

**Theorem 3.9** (Consequence of the Closed Graph Theorem)**.** *If $A$ is closed and $D(A) = X$ then $A$ is bounded.*

**Definition 3.10** (Definition of Closable Operator)**.** *We say that $A$ is* ***closable*** *when $\overline{G(A)}$ is the graph of an operator $(\overline{A}, D(\overline{A}))$. $\overline{A}$ is called the* ***closure*** *of $A$.*

An example of Non-closable operator without giving details.

**Remark 3.11** (An example of Non-closable operator)**.** *An example of Non-closable operator can be given in Hilbert spaces when we discuss the adjoint operator*

We use the following theorem:

**Theorem 3.12.** *The continuous functions $C([a, b])$ form a dense subset of $L^2([a, b])$ (page 327 in Rudin).*

**Example 3.13.** *Consider the $L^2([0, 1])$ and the dense subspace*

- *$D = C([0, 1])$.*
- *Define $A$ on $D$ by $A(f) = f(0)$.*

*Then $A$ is not closable.*

This is a densely defined operator, but one can prove that its adjoint is not densely defined.

Since for the class of densely defined operators we have that

- a closed linear extension exists if and only if
- the adjoint is densely defined this operator.

So $A$ is not closable.

**Proposition 3.14.** *Let $(A, D(A))$ be closed and define the graph norm on $D(A)$ by*

$$||x||_{D(A)} := ||x||_X + ||Ax||_Y.$$

*Then $(D(A), ||\cdot||_{D(A)})$ is a Banach space and $A : D(A) \subset X \to Y$ is continuous with respect to this norm.*

## 4. Resolvent set, the spectrum and resolvent operator

We introduce the crucial concepts of resolvent set, the spectrum and resolvent operator.

**Definition 4.1** (Resolvent set, spectrum and resolvent operator)**.** *Let $(A, D(A))$ be a Closed operator on $X$, i.e. $X = Y$, where $X$ is a Banach space over the complex numbers $\mathbb{C}$.*

- *The set*
$$\rho(A) := \{z \in \mathbb{C} \ : \ A - z : D(A) \to X \quad \text{is bijective,} \\ \text{and } (A-z)^{-1} \in \mathcal{B}(X)\}$$
*is called the resolvent set of $A$.*
- *The complement $\sigma(A) := \mathbb{C} \setminus \rho(A)$ is the spectrum of $A$.*
- *For $z \in \rho(A)$ the operator*
$$R(z, A) := (A - z)^{-1}$$
*is called the resolvent operator of $A$.*

**Definition 4.2** (Decomposition of the Spectrum $\sigma(A)$)**.** *For the spectrum $\sigma(A)$ we have the decomposition:*

$$\sigma(A) = \sigma_p(A) \bigcup \sigma_c(A) \bigcup \sigma_r(A),$$

*where*

- *The point spectrum:*
$$\sigma_p(A) := \{z \in \mathbb{C} \ : \ A - z \text{ is not one-to-one}\}.$$
- *The continuous spectrum:*
$$\sigma_c(A) := \{z \in \mathbb{C} \ : \ A - z \text{ is one-to-one,} \\ \text{with } \operatorname{range}(A-z) \neq X \quad \text{but} \quad \overline{\operatorname{range}(A-z)} = X\}$$
- *The residual spectrum:*
$$\sigma_r(A) := \{z \in \mathbb{C} \ : \ A - z \text{ is one-to-one, but } \overline{\operatorname{range}(A-z)} \neq X\}.$$

**Remark 4.3** (Remark on $\sigma_c(A)$ and $\sigma_r(A)$)**.** *Observe that: the resolvent operator $(A-z)^{-1}$ has a domain and is not defined on all $X$.*

- *By definition of the continuous spectrum:*

$$\sigma_c(A) := \{z \in \mathbb{C} \ : \ A - z \text{ is one-to-one,}$$
$$\text{with } \operatorname{range}(A-z) \neq X \quad \text{but} \quad \overline{\operatorname{range}(A-z)} = X\Big\}$$

- *since:*

$$D((A-z)^{-1}) = \operatorname{range}(A-z) \neq X, \ \text{but} \ \ \overline{D((A-z)^{-1})} = X$$

- *The residual spectrum is defined as:*

$$\sigma_r(A) := \Big\{z \in \mathbb{C} \ : \ A - z \text{ is one-to-one, but } \overline{\operatorname{range}(A-z)} \neq X\Big\}$$

- *since:*

$$D((A-z)^{-1}) = \operatorname{range}(A-z) \neq X, \ \text{and} \ \ \overline{D((A-z)^{-1})} \neq X$$

**Theorem 4.4** (Linear Operators on Finite-Dimensional Spaces)**.** *Let $X$ be a finite-dimensional vector space over the field $\mathbb{F}$, and let $A : X \to X$ be a linear operator. Then $A$ is injective if and only if $A$ is surjective.*

- $\dim(\ker(A)) = 0$ if and only if $\dim(\operatorname{image}(A)) = \dim(X)$.
- In finite-dimensional spaces, the rank-nullity theorem

$$\dim(Im(A)) + \dim(Ker(A)) = \dim(X)$$

  ensures that injectivity and surjectivity are equivalent for linear operators.
- In this case there are just the eigenvalues, that is $\sigma_p(A)$.

## 5. The resolvent equation

**Theorem 5.1.** *Let $(A, D(A))$ be a closed operator.*

(I) *The resolvent set $\rho(A)$ is open and the spectrum $\sigma(A)$ is closed.*
(II) *The resolvent operator $R(z, A)$ defines an analytic function $\rho(A) \to \mathcal{B}(X)$.*
(III) *For $z, w \in \rho(A)$ we have*

$$R(z, A) - R(w, A) = (z - w)R(z, A)R(w, A), \tag{5.1}$$

*in particular $R(z, A)$ and $R(w, A)$ commute, i.e.*

$$R(z, A)R(w, A) = R(w, A)R(z, A).$$

*(I): $\rho(A)$ is Open (as a Consequence, $\sigma(A)$ is Closed).* Let $z_0 \in \rho(A)$. We need to show that there exists an open neighborhood of $z_0$ contained in $\rho(A)$. Since $(A - z_0)^{-1}$ is bounded, there exists $M > 0$ such that $\|(A - z_0)^{-1}\| \leq M$.

Tale $z \in \mathbb{C}$ and consider the operator

$$A - z = A - z + z_0 - z_0 = A - z_0 - z + z_0 = (A - z_0)\Big(I - (z - z_0)(A - z_0)^{-1}\Big)$$

now we look at the operator

$$I - (z - z_0)(A - z_0)^{-1}.$$

For $|z - z_0| < 1/M$, we have

$$\|(z - z_0)(A - z_0)^{-1}\| \leq |z - z_0|\|(A - z_0)^{-1}\| < 1.$$

Therefore, by the Neumann series, the operator

$$I - (z - z_0)(A - z_0)^{-1}$$

is invertible for $|z - z_0| < 1/M$. Consequently, also $(A - z)$ is also invertible since

$$(A - z) = (A - z_0)(I - (z - z_0)(A - z_0)^{-1}).$$

This shows that the open ball centered at $z_0$ with radius $1/M$ is contained in $\rho(A)$. Hence, $\rho(A)$ is open and the spectrum, as a consequence, is closed. □

*(II) Analyticity of $R(z, A)$.* Fix $z_0 \in \rho(A)$. For any $z$ in a small neighborhood of $z_0$, i.e., for $|z - z_0| < 1/M$ the operator $(A - z)$ is invertible and we have

$$(A - z)^{-1} = (A - z_0)^{-1}\Big(I - (z - z_0)(A - z_0)^{-1}\Big)^{-1}.$$

Using the Neumann series, we have

$$\Big(I - (z - z_0)(A - z_0)^{-1}\Big)^{-1} = \sum_{n=0}^{\infty}(z_0 - z)^n[(A - z_0)^{-1}]^n.$$

and we finally have

$$(A - z)^{-1} = (A - z_0)^{-1}\sum_{n=0}^{\infty}(z_0 - z)^n[(A - z_0)^{-1}]^n = \sum_{n=0}^{\infty}(A - z_0)^{-1-n}(z_0 - z)^n.$$

This series converges for $|z - z_0| < 1/M$. Therefore, $R(z, A)$ is an analytic function of $z$ in a neighborhood of $z_0$. Since $z_0$ was arbitrary in $\rho(A)$, $R(z, A)$ is analytic on $\rho(A)$.

□

*(III) Resolvent Identity.* For $z, w \in \rho(A)$, recall that by definition

$$(A - z)(A - z)^{-1} = I_X \quad \text{and} \quad (A - w)(A - w)^{-1} = I_X$$
$$(A - z)^{-1}(A - z) = I_{D(A)} \quad \text{and} \quad (A - w)^{-1}(A - w) = I_{D(A)}$$

we can write

$$\begin{aligned} R(z, A) - R(w, A) &= (A - z)^{-1} - (A - w)^{-1} \\ &= (A - z)^{-1}[(A - w) - (A - z)](A - w)^{-1} \\ &= (A - z)^{-1}[-w + z](A - w)^{-1} \\ &= (z - w)R(z, A)R(w, A) \end{aligned}$$

The resolvent identity immediately implies that $R(z, A)$ and $R(w, A)$ commute for any $z, w \in \rho(A)$. □

**Remark 5.2.** (I) *If $A \in \mathcal{B}(X)$ then $\sigma(A) \neq 0$ (We will see this fact). However, there are unbounded operators with empty spectrum, and thus $\sigma(A) = 0$ can occur.*

(II) *If* $\rho(A) \neq 0$ *then* $A$ *is closed, since* $R(z, A) \in \mathcal{B}(X)$ *as a consequence* $(A - z)^{-1}$ *is closed so the inverse* $A - z$ *is closed and so* $A$ *is also closed.*